Basics of FE-Analysis
BK10A6400
Lecture 1: Introduction

Marko Matikainen

LUT University, Finland

August 31, 2026

General information

Topic: Displacement-based linear finite element method for problems in structural and solid mechanics.

Lectures: Marko Matikainen (Check times from teaching schedule. Often on Mondays in a class 7343)

Exercises: Uzair Shakeel and Samar Keshavarz (every second week, but if needed, every week).

Course Schedule

Week	Date	Time	Room	Topic
36	31.8.2026	14:15–16:00	7343.1+7343.2	Introduction to finite element analysis
37	7.9.2026	14:15–16:00	7343.1+7343.2	?
38	14.9.2026	14:15–16:00	7343.1+7343.2	?
39	**23.9.2026**	**08:15–10:00**	**1316 Auditorium**	?
40	28.9.2026	14:15–16:00	7343.1+7343.2	?
41	5.10.2026	14:15–16:00	7343.1+7343.2	?
42	12.10.2026	14:15–16:00	7343.1+7343.2	?

Exception

The lecture on **Wednesday 23 September** is held at **08:15–10:00** in **Auditorium 1316**.

Course schedule – continued

Week	Date	Time	Room	Topic
44	26.10.2026	14:15–16:00	7343.1+7343.2	?
45	2.11.2026	14:15–16:00	7343.1+7343.2	?
46	9.11.2026	14:15–16:00	7343.1+7343.2	?
47	16.11.2026	14:15–16:00	7343.1+7343.2	?
48	23.11.2026	14:15–16:00	7343.1+7343.2	?
49	30.11.2026	14:15–16:00	7343.1+7343.2	?
50	7.12.2026	14:15–16:00	7343.1+7343.2	?

General schedule

Lectures are normally held on Mondays at **14:15–16:00** in room **7343.1+7343.2**.

Exercises

Week	Date	Time	Room
38	16.9.2026	16:15–18:00	6520 Computer Classroom
40	30.9.2026	16:15–18:00	6520 Computer Classroom
42	14.10.2026	16:15–18:00	6520 Computer Classroom
44	28.10.2026	16:15–18:00	6520 Computer Classroom
46	11.11.2026	16:15–18:00	6520 Computer Classroom
48	25.11.2026	16:15–18:00	6520 Computer Classroom

Exercise schedule

Exercises are held every second Wednesday at **16:00–18:00** in room **6520**.

Course assignments and points

The course includes

- **4 exercise assignments**
  - maximum **10 points** from each assignment,
  - maximum **40 points** in total,
- approximately **6–8 bonus tasks**
  - typically **3–5 points** from each bonus task,
  - the number of points depends on the task.

Passing the course:

- At least **50 points** are required to pass the course.
- The nominal maximum is **100 points**, but the actual maximum can be higher because of the bonus tasks.
- The **exam is not mandatory** if you collect enough points from the other course activities.
- In practice, it is usually possible to pass the course without taking the exam.

Software choices during the course

At the beginning of the course, choose

FE software

- FEMAP
- ANSYS

Programming environment

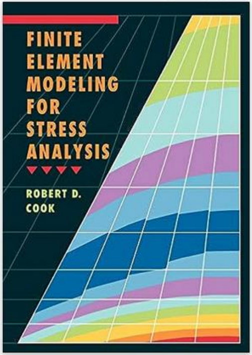
- MATLAB
- Python

Use your selected tools consistently throughout the course.

This allows you to concentrate on learning finite element analysis instead of repeatedly learning new software.

Course material

- Robert D. Cook: "Finite Element Modeling for Stress Analysis" or "Concepts and Applications of Finite Element Analysis"
- Matti K. Hakala : "Lujuusopin elementtimenetelmä"
- Svein Linge and Hans Petter Langtangen : "Programming for Computations - MATLAB/Octave" Link
- Claus Führer, Jan Erik Solem and Olivier Verdier: "Computing with Python"
- MATLAB - codes and Python - codes (hopefully)
- Some ANSYS macros
- Some lecture notes (but not hundreds of slides) will be available.
- Learning by doing...also during lectures!
- It is essential to attend the lectures. Guidance for the bonus tasks.
- It is important to participate in the exercises. Guidance on the assignments.


FINITE
ELEMENT
MODELING
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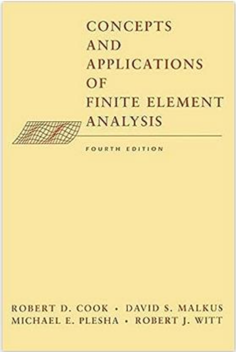
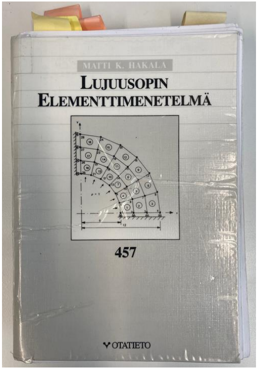
ROBERT D.
COOK


CONCEPTS
AND
APPLICATIONS
OF
FINITE ELEMENT
ANALYSIS
FOURTH EDITION
ROBERT D. COOK • DAVID S. MALKUS
MICHAEL E. PLESHA • ROBERT J. WITT


MATTI K. HAKALA
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Preliminary requirements

- From the third year onwards.
- Ready to work and learn mathematical-oriented (engineering) stuff. The more time you invest, the more you learn.
- Strength of materials
- Matrix calculus
- Anyone used MATLAB or Python?

Passing the course

- 4 credits (104 hours work)
- Course grade: 0-5, exercises 40% and exam 60%.
- Four exercises a' 10 points
- Exam includes multiple choice questions (60 points).
- Attendance for contact hours is strongly recommended. Be on time!
- During lectures, bonus tasks will be shared and guidance will be given.
- Bonus points (from bonus tasks) will be included in exam points.
- You should earn 50 points to pass the course.
- It is possible to earn more than 100 points due to bonus tasks, but the grade cannot be higher than 5 :-D.

Passing the course

Bonus tasks

Have to return via Moodle before the deadline. No points after it. Usually around a week to do it.

Passing the course

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Exam

It is possible to pass the course without an exam by earning 50 points.

Working load

- 4 ECT $\cdot$ 26h = 104 h
- Contact hours (lectures + exercises): 14 weeks $\cdot$ 2h + 6 weeks $\cdot$ 2h = 40h
- 50 h for all assignments and exam
- $\approx$ 3.5 hours per week (without contact hours).

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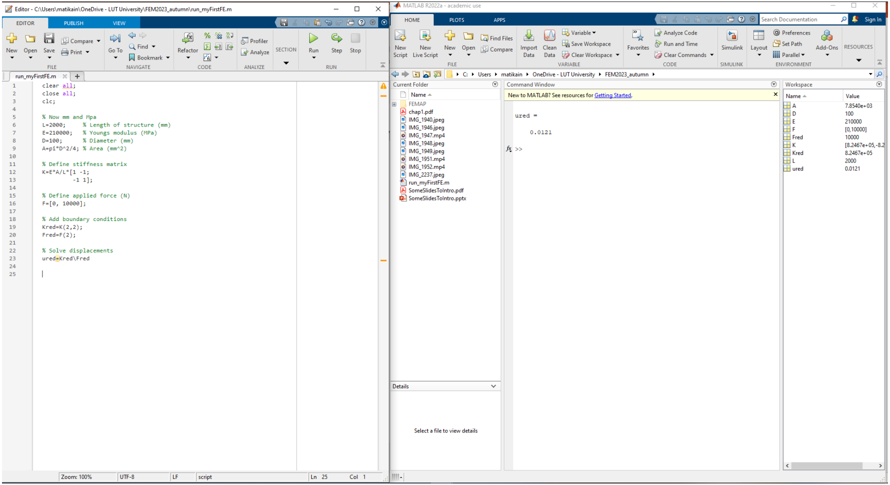
Average student's working load excluding contact hours

$$\approx 3.5 \text{ hours per week}$$

Bonus tasks

- Solving small problems in structural and solid mechanics with FEM using MATLAB
- You must use the code templates shared in the course.
- Everybody has own initial parameters for the problem.

myFirstFE in MATLAB



After the course

After having passed the course, the student can:

- understand the basic theoretical background of the displacement-based FEM
- solve displacements (also strains and stresses) about statically loaded mechanical structures.
- solve eigenfrequencies and modes of mechanical structures.
- use FE-analysis software for simple mechanical structure analysis

Goal of lectures

The purpose of the lectures is to give basic knowledge about

- elemental stiffness matrix
- assembling the global stiffness matrix
- boundary conditions and loads
- solution and source of errors
- linear static and linearized dynamics analysis with FEM

FE analysis using the commercial finite element software FEMAP will be introduced in the exercises.

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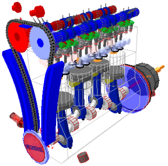
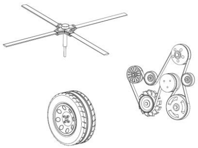
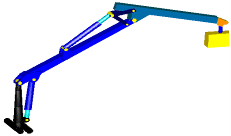
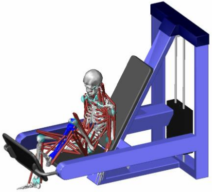
Motivation

- Displacements, strains and stresses in structures during static and/or dynamical loading
- Gain a better understanding of loadings that lead to too large deflections or even failure.









Motivation

Why should you know the FEM deeper than just being a dummy user of FEA packages?

- to know the limitations of finite elements.
- to understand the source of errors.
- to choose a proper solution method (linear, nonlinear, time-dependent,..) for your problem
- to feel great after understanding difficult stuff.

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The most important thing to learn FEM is to know

the right way to set up your problem.

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Warning!

Computer programs follow the philosophy of GIGO!

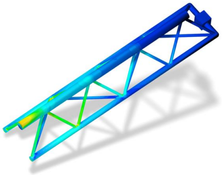
What is FEM?

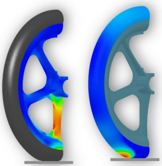
Link

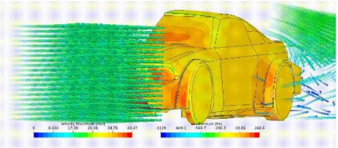
- Generally: Numerical method to solve a set of partial differential equations.
- In the course: Numerical method to solve displacements of structures.
- Often an important part of multiphysics software.

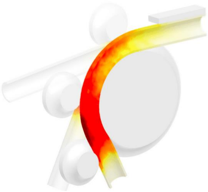
Link

Examples from SimScape https://www.simscale.com/docs/tutorials/









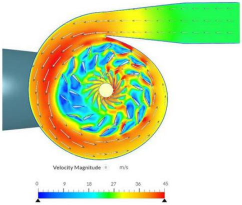
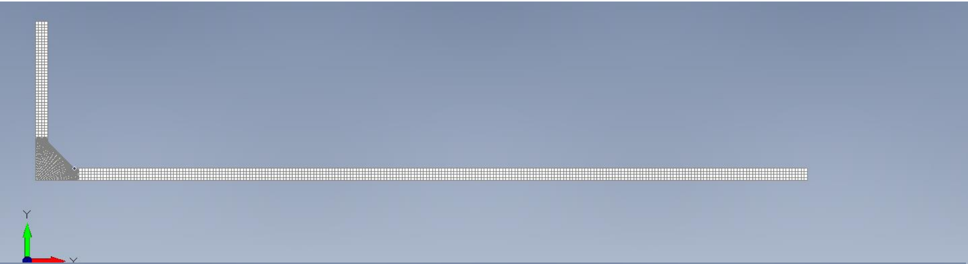


Figure 1: Velocity vectors plotted onto the velocity field.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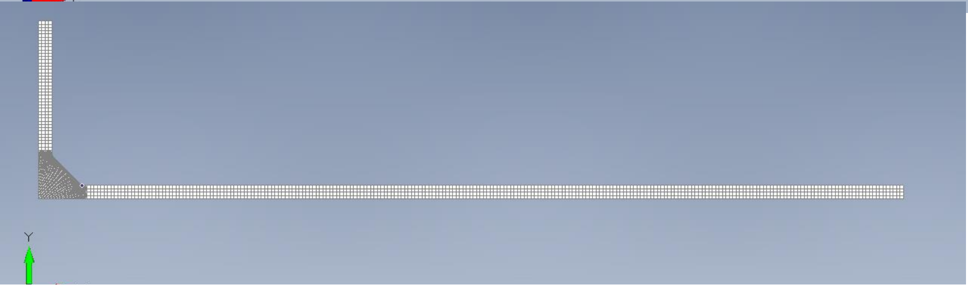
X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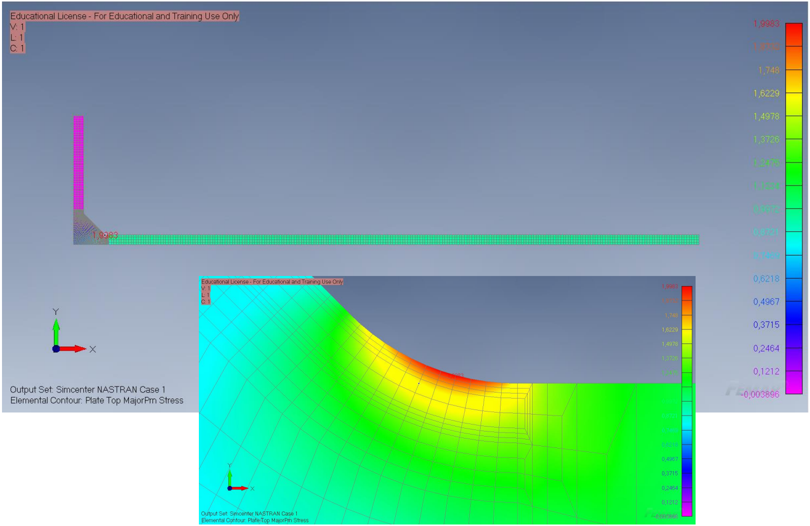

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-0,003896
Y
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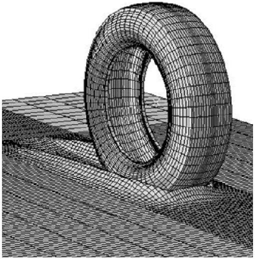
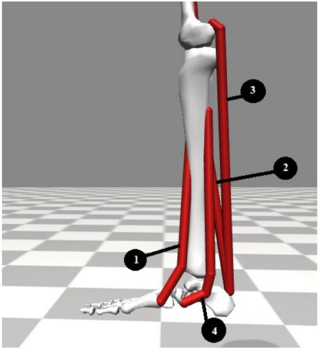
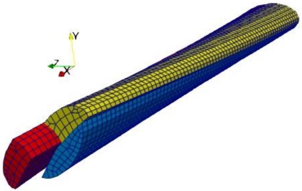
Output Set: Simcenter NASTRAN Case 1
Elemental Contour: Plate Top MajorPrn Stress





Muscles:

1. Tibialis anterior	TA
2. Soleus	SOL
3. Gastrocnemius	GA
4. Tibialis posterior	TP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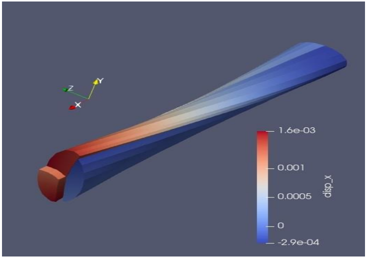
Y
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1.6e-03
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disp_x

Static, implicit and explicit FE analysis

First separate two different questions:

1. What kind of analysis?

- **Static**
- **Dynamic**

2. How is time integration done?

- **Implicit**
- **Explicit**

Important

Static, implicit and explicit are not three parallel alternatives.
Implicit and explicit refer mainly to how a **time-dependent problem** is advanced from one time step to the next.

1. Static FE analysis

In static analysis, inertia is neglected and we search for an **equilibrium configuration.**

For a linear FE model:

$$\mathbf{K}\mathbf{u} = \mathbf{f}$$

stiffness · displacement = external load

For a single spring:

$$ku = f \qquad \Longrightarrow \qquad u = \frac{f}{k}$$

Physical interpretation

The structure is in equilibrium under the applied load. There is no need to follow its motion in time.

For a nonlinear static problem:

$$\mathbf{f}_{\mathrm{int}}(\mathbf{u}) = \mathbf{f}_{\mathrm{ext}}$$

and the equilibrium is usually found iteratively, e.g. with

2. Dynamic FE analysis

If inertia matters, the motion of the structure must be followed in time.

$$\mathbf{M}\ddot{\mathbf{u}} + \mathbf{C}\dot{\mathbf{u}} + \mathbf{K}\mathbf{u} = \mathbf{f}(t)$$

mass + damping + stiffness = time-dependent loading

For one degree of freedom:

$$m\ddot{u} + c\dot{u} + ku = f(t)$$

Now the solution is not only one displacement vector:

$$\mathbf{u}(t_0),\ \mathbf{u}(t_1),\ \mathbf{u}(t_2),\ldots$$

The key question

How do we calculate the solution at t_{n+1} from the already known solution at t_n?

This is where **implicit** and **explicit** time integration differ.

3. Explicit time integration

Consider the simplest undamped equation:

$$m\ddot{u} + ku = f(t)$$

Approximate acceleration at the **known** time t_n:

$$\ddot{u}_n \approx \frac{u_{n+1} - 2u_n + u_{n-1}}{\Delta t^2}$$

Substitution gives

$$\boxed{u_{n+1} = 2u_n - u_{n-1} + \frac{\Delta t^2}{m}(f_n - ku_n)}$$

Why is this explicit?

Everything on the right-hand side is already known.

$$\boxed{\text{Known state} \implies \text{directly calculate next state}}$$

In FE explicit solvers, the mass matrix is usually **lumped**:

$$\mathbf{M}_L = \mathrm{diag}(m_1, m_2, \ldots, m_N)$$

4. Implicit time integration

Again consider

$$m\ddot{u} + ku = f(t)$$

In an implicit method, equilibrium is enforced at the **new time** t_{n+1}:

$$m\ddot{u}_{n+1} + ku_{n+1} = f_{n+1}$$

But both u_{n+1} and $\ddot{u}_{n+1}$ contain the **unknown future state.** After time discretization, we obtain a system of the form

$$\boxed{\mathbf{K}_{\text{eff}}\,\mathbf{u}_{n+1} = \mathbf{f}_{\text{eff}}}$$

Why is this implicit?

The unknown future state appears inside the equation that must be solved.

$$\boxed{\text{Form equations at } t_{n+1} \implies \text{solve for the new state}}$$

For nonlinear problems, this solution usually requires iterations at each time step.

Static vs. implicit vs. explicit: the main idea

	Static	**Implicit dynamics**	**Explicit dynamics**
Main equation	$\mathbf{Ku} = \mathbf{f}$	equilibrium at t_{n+1}	equilibrium at t_n
Time	not followed	step-by-step	step-by-step
At each step	solve equilibrium	solve system / iterate	direct update
Typical time step	–	relatively large	small
Typical applications	static loading	structural dynamics, nonlinear analysis	impact, crash, severe contact

Remember

Static: solve equilibrium Implicit: solve the next state Explicit: calcul

How this course starts

Most of this course begins from the classical **displacement-based static finite element method**:

$$\boxed{\mathbf{Ku} = \mathbf{f}}$$

Once we understand

elements $\rightarrow$ stiffness matrices $\rightarrow$ assembly $\rightarrow$ boundary condition

the same FE ideas can be extended to

$$\boxed{\text{nonlinear statics} \quad + \quad \text{implicit dynamics} \quad + \quad \text{explicit dynamics}}$$

Main message

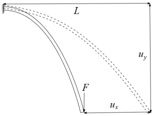
The finite element discretization is the foundation. The solver and time-integration method determine how the resulting equations are solved.

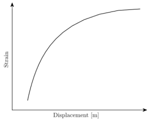
FEM

- Numerical (approximative) method for solving partial differential equations (PDE).
- Problems in structural analysis, fluid flow, electromagnetic etc.
- Originally developed by engineers for structural analysis in Aerospace engineering in 1950's.
- The idea is to divide the complicated structure (domain) into a number of simple geometrical elements.
- The primary field (such as displacement) of a single element is interpolated between nodes using simple polynomials.
- Elements are usually connected via common nodes.
- In structural analysis, primary field variables can be (nodal) displacements, stresses
- Displacement-based FEM

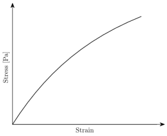
$$\boldsymbol{u} = \boldsymbol{K}^{-1}\boldsymbol{f}_{ext}$$

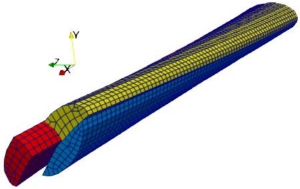
Geometric: Nonlinear strain-displacement



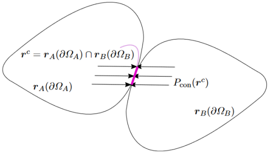


Material: Nonlinear stress-strain



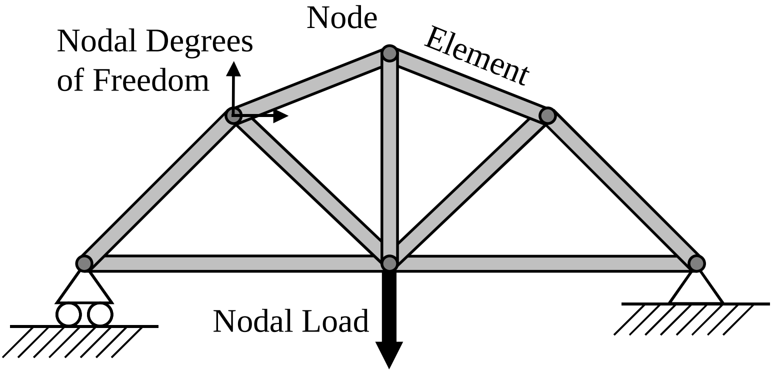


Contact: Nonlinear constraint over boundary



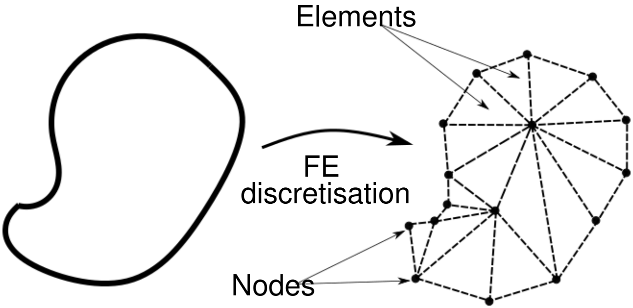
Schematic FE discretisation of an arbitrary 1D structure

- Rod (truss) -type structure
- Beam-type structure



$$\boldsymbol{u} = \boldsymbol{K}^{-1}\boldsymbol{f}_{ext}$$

Schematic FE discretisation of an arbitrary 2D structure



$$\boldsymbol{u} = \boldsymbol{K}^{-1}\boldsymbol{f}_{ext}$$

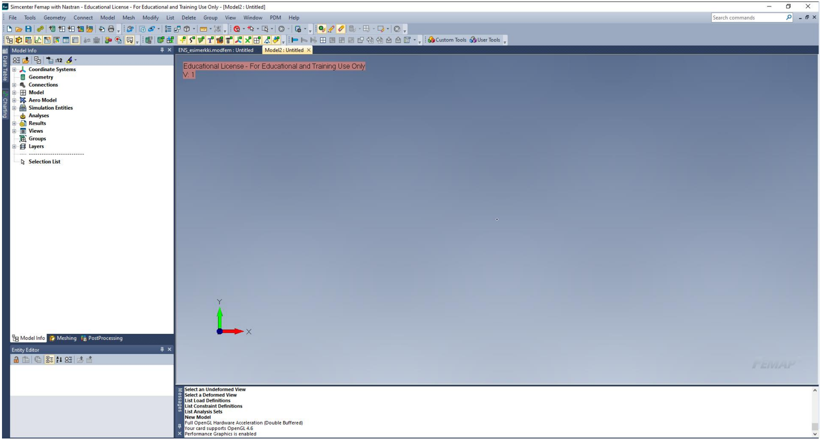
More applications

Link

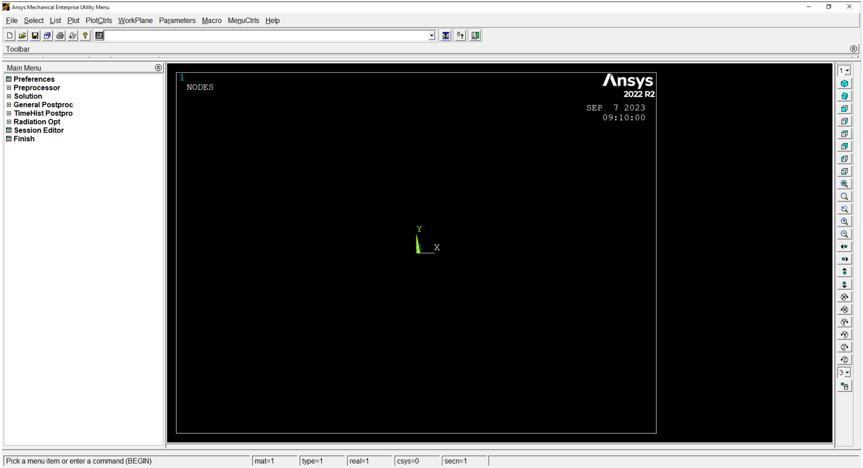
FE softwares

- Preprocessor
  - Element type
  - Create mesh
  - Assembling
  - Material
  - Boundary conditions
  - Loading
- Solver
  - Solution type
- Postprocessor
  - Visualization of displacement, strains and stresses
- Nowadays, many commercial FE softwares (ANSYS, ABAQUS, SimScale, Nastran etc) exist as well as open source FEA packages such as Elmer (developed by Finnish Ministry of Education's CSC) and Salome Meca/Code Aster.
- A commercial FE software FEMAP (pre- and post-processing) is used in exercises. It uses NX/Nastran as solver.

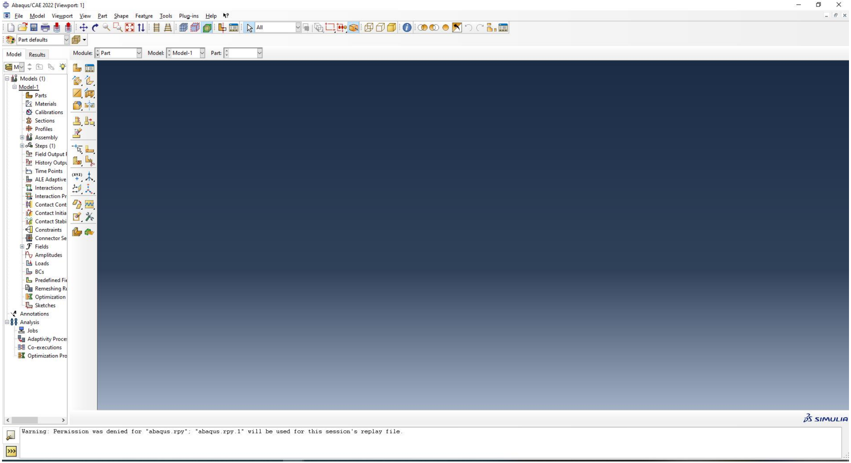
FEMAP graphical user interface



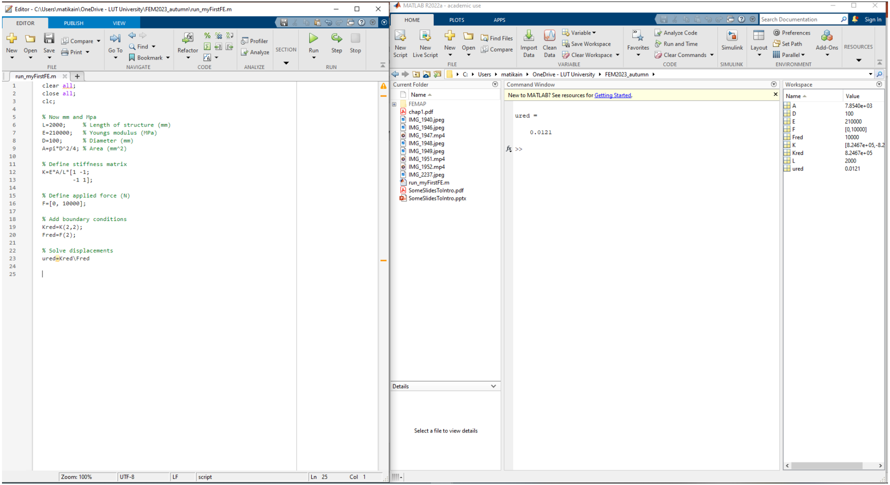
ANSYS graphical user interface



ABAQUS graphical user interface



myFirstFE in MATLAB



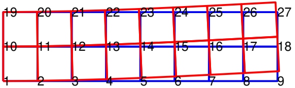
Needs for the numerical solution method

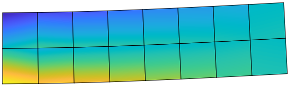
Any numerical solution method for engineering problems should be reliable and efficient.

- Applicable
- Accurate
- Stable
- Inexpensive

Mesh, elements, nodes and degrees of freedom

- Degrees of Freedom, element, nodes, nodal displacements and displacement field.
- Elements are interconnected via nodes.
- There exist hundreds of different element types.
- The most common element types in the structural analysis are rod (truss), beam, shell and solid elements.
- As an example, a simple cantilever problem is analyzed by a four-node plane elements.





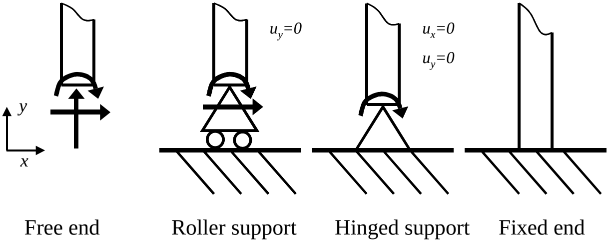
Boundary conditions: constraints and loadings

- If bc:s are not defined correctly, the solution is not a solution for the problem, or the solution cannot be found.
- Loadings in displacement-based FE can be force, pressure, and temperature.
- BC:s and loadings can be applied to points, edges, and surfaces of the objects, but they are always distributed on nodes in the code.

Boundary conditions

The degrees of freedom (DOF) can be divided into translations and rotations. DOF's can be either free or fixed.

- Free end
- Roller support
- Hinged support
- Fixed end



Recap on the basics of Strength of Materials (one-dimensional case)

Strain:

$$\varepsilon = \frac{\delta}{L}$$

Displacement:

$$\delta = L_1 - L$$

Stress (Hooke's law):

$$\sigma = E\varepsilon$$

Stress can also be defined as:

$$\sigma = \frac{F}{A}$$

Substitute strain and stress into Hooke's law:

$$\delta = \frac{FL}{EA}$$

The procedure of linear FEM

1. Formulation of elemental stiffness matrices
   - Derive (or use ready solution) local elemental stiffness matrix $\overline{\boldsymbol{K}}$
2. Assembling global stiffness matrix and a vector of external (global) forces for a structure.
   - Compute global elemental stiffness matrix for element k: $\boldsymbol{K}^{(k)} = \boldsymbol{T}^T \overline{\boldsymbol{K}}^{(k)} \boldsymbol{T}$.
   - Assembling global stiffness matrix $\boldsymbol{K}_{sys}$
   - External force vector of a structure $\boldsymbol{f}$
3. Boundary conditions (and global force vector): $\boldsymbol{K}_c, \boldsymbol{f}_c$
4. Solving unknown (nodal displacements) from a system of linear equations $\boldsymbol{u}_c = \boldsymbol{K}_c^{-1} \boldsymbol{f}_c$
   - Inverse of stiffness matrix!
   - Direct solvers
   - Iterative solvers
5. Solving forces, strains and stresses. Visualizations.

Stiffness matrix of a rod/truss element

$$\overline{\boldsymbol{K}} = \frac{EA}{L} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \tag{1}$$

- Symmetric
- Determinant: $\det \overline{K} = 0$
- A stiffness matrix of a rod element has one rigid body mode (in axial direction).
- Matrix rank: $\operatorname{rank} \overline{K} = 1$

MATLAB: Solve displacements of a simple rod under axial loading using one rod (truss) element only.